

Olist E-Commerce

Customer Retention Analysis

*What separates one-time buyers from repeat customers,
and what should Olist do about it?*

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Dataset: Olist Brazilian E-Commerce (99,441 orders · 93,358 unique customers)

Executive Summary

Business Question: What separates one-time buyers from repeat customers, and where should Olist focus its retention efforts?

Finding: About 97% of customers placed only one observed order during the dataset period, while 2,801 customers made more than one purchase.

Repeat and one-time customers had very similar product preferences and review scores. Four of the top five product categories overlapped between the two groups, while average review scores were 4.15 for one-time buyers and 4.21 for repeat customers.

Among customers who returned, the average time between the first and second purchase was 81.2 days. The analysis also found large differences in repeat purchase rates across sellers.

Why It Matters: Repeat customers spent an average of R\$308.53 per customer, compared with R\$160.74 for one-time buyers during the observed period.

Although repeat customers represented only a small share of the customer base, their average observed spend was approximately 92% higher.

This makes repeat purchasing an important area for further investigation and testing.

Recommendation: Olist should test customer re-engagement before the average second-purchase point. A campaign between days 55 and 65 after the first delivered order could be used as an initial experiment and measured against a control group.

The company should also review sellers with unusually high repeat purchase rates to identify operational differences that were not available in the current dataset.

Method: I used MySQL to segment **93,358 customers** by purchase frequency and compare revenue, product categories, review scores, purchase timing, and seller-level repeat rates.

Six SQL queries using CTEs, window functions, conditional aggregation, and multi-table joins were used to narrow the analysis from overall purchase frequency to the timing and seller patterns associated with repeat purchasing.

Investigation

1. How many customers ever come back?

The first step was establishing the shape of the retention problem: across all delivered orders, how many distinct customers placed 1 order, 2 orders, 3 orders, and so on.

Orders Placed	Customers	% of Customers
1	90,557	97.00%
2	2,573	2.76%
3	181	0.19%
4+	47	0.05%

What this tells us: 97 out of every 100 customers in the dataset placed only one observed order. Only 3% made more than one purchase during the period covered by the data. This shows that repeat purchasing was uncommon and makes customer retention an important area for further investigation.

2. Do repeat customers actually matter financially?

Before recommending any investment in retention, the next question was whether the rare repeat customers are worth the effort, measured in revenue contribution and average spend per person.

Customer Type	Customers	Total Revenue	Avg. Spend
One-Time	90,557	R\$14.56M	R\$160.74
Repeat	2,801	R\$864K	R\$308.53

What this tells us: Repeat customers spent 92% more per person during the observed period than one-time buyers (R\$308.53 vs R\$160.74). Although repeat customers represented only 2.8% of the customer base, their higher average spend shows why increasing repeat purchasing could have a meaningful revenue impact.

3. Do repeat customers buy different products?

If repeat customers concentrate in specific categories, that would point toward a product-led fix. The top 5 categories by revenue were compared between the two groups.

Rank	One-Time Top Category	Rank	Repeat Top Category
1	beleza_saude (health/beauty)	1	cama_mesa_banho (home/bath)
2	relogios_presentes (watches/gifts)	2	esporte_lazer (sports/leisure)

3	cama_mesa_banho (home/bath)	3	moveis_decoracao (furniture/decor)
4	esporte_lazer (sports/leisure)	4	informatica_acessorios (computer)
5	informatica_acessorios (computer)	5	beleza_saude (health/beauty)

What this tells us: Four of the top five product categories appear in both customer groups, although the ranking differs. Repeat and one-time customers therefore show broadly similar category preferences. Based on this comparison, product category alone does not appear to explain why some customers make another purchase.

4. Are one-time buyers leaving because of a bad experience?

After comparing product categories, I examined review scores to see whether one-time buyers reported a noticeably worse experience than repeat customers.

Customer Type	Customers	Reviews	Avg. Score
One-Time	89,963	90,053	4.15 / 5
Repeat	2,792	5,594	4.21 / 5

What this tells us: Average review scores are very similar between the two groups: 4.15 for one-time buyers and 4.21 for repeat customers. The available review data does not show a major satisfaction gap between the groups. This suggests that review score alone is unlikely to explain the low level of repeat purchasing.

5. For customers who do return, how long does it take?

With product and service ruled out, the final question was timing: how many days pass between a customer's first and second purchase, for those who do come back?

Purchase Number	Customers Reaching It	Avg. Days Since Last
2nd purchase	2,801	81.2 days
3rd purchase	228	62.8 days
4th purchase	47	60.6 days
5th purchase	19	61.2 days

What this tells us: Customers who made a second purchase took an average of 81.2 days to do so. Among customers who continued purchasing, the average gaps before the third, fourth, and fifth purchases were shorter at approximately 61 to 63 days. This makes the period before the average second purchase a useful window for testing re-engagement. A campaign around days 55 to 65 could be tested against a control group to measure whether earlier contact increases second-purchase rates.

6. Secondary Finding: Does Retention Vary by Seller?

After finding that category preferences and review scores were similar between one-time and repeat customers, I examined whether repeat purchase rates varied by seller. Sellers with at least 50 customers were ranked by the share of their customers who made more than one purchase during the observed period.

Seller City	State	Customers	Repeat	Repeat %
Sao Jose dos Campos	SP	88	20	22.73%
Sao Paulo	SP	58	13	22.41%
Porto Ferreira	SP	68	12	17.65%
Timbo	SC	102	15	14.71%
Sete Lagoas	MG	219	29	13.24%
Platform Average	-----	93,358	2,801	2.76%

What this tells us: Repeat purchase rates varied considerably among the sellers included in the analysis. The highest observed rate was 22.73%, compared with a platform average of 2.76%. The top sellers were located across several Brazilian states, so no clear location pattern was identified from this comparison. These differences make seller-level performance worth further investigation, but the current dataset does not explain what caused the higher repeat rates. Delivery performance, pricing, promotions, communication, or other seller practices would need to be examined separately.

Recommendation

The analysis identified two areas worth testing: the timing of customer re-engagement and the differences in repeat purchase rates across sellers. The recommendations below are based on the observed patterns, but should be tested before wider implementation.

1. Test a Day 55 - 65 Reactivation Campaign

Customers who made a second purchase returned after an average of 81.2 days.

Olist should test an automated re-engagement message between days 55 and 65 after a customer's first delivered order. The campaign could use a personalized product recommendation or a limited incentive to encourage another purchase.

The 55 - 65 day period should be treated as a starting point for an experiment, not a proven optimal window. Results should be compared with a control group and measured using second-purchase conversion rate and incremental revenue.

2. Study Sellers With Unusually High Repeat Purchase Rates

Some sellers recorded repeat purchase rates considerably above the platform average of 2.76%.

Olist should compare high-repeat sellers with sellers serving a similar number of customers but recording lower repeat rates. The review should examine delivery performance, pricing, promotions, product mix, and other seller-level differences available in the data.

Where additional operational data is available, packaging, seller communication, and post-purchase follow-up should also be investigated.

The goal is to identify whether any repeatable seller practices are associated with higher repeat purchasing.

3. Investigate Additional Causes of Low Repeat Purchasing

Similar review scores and top category preferences suggest that these two factors alone do not explain the difference between one-time and repeat customers.

Further analysis should examine delivery times, late deliveries, cancellations, refunds, payment behaviour, freight costs, and customer location. Customer acquisition source and post-purchase communication should also be reviewed if those data are available.

This would help separate factors associated with repeat purchasing from patterns that only appear similar between the two customer groups.

4. Measure Retention Using Customer Cohorts

The current analysis covers customers observed within a fixed dataset period. Customers who made their first purchase near the end of the dataset had less time to make another purchase than customers acquired earlier.

Future retention analysis should group customers by first-purchase month and give each cohort a consistent observation window.

This would provide a clearer measure of repeat purchase behaviour and help determine whether retention improved or declined over time.

Methodology

Dataset

- Olist Brazilian E-Commerce Public Dataset (Kaggle)
- 99,441 delivered orders across 93,358 unique customers
- Tables used: customers, orders, order_items, order_reviews, products

Tools

- MySQL Workbench - data import, cleaning, and all six analytical queries
- Python (pandas, SQLAlchemy) - CSV-to-MySQL import with explicit data typing and indexing for query performance

Techniques Used

- Common Table Expressions (CTEs) for multi-step, readable query logic
- Window functions - RANK(), ROW_NUMBER(), and LAG() - for ranking categories and measuring time between purchases
- Multi-table JOINS across customers, orders, order_items, order_reviews, and products

- CASE WHEN segmentation to classify customers as One-Time or Repeat
- DATEDIFF() combined with LAG() to calculate days between consecutive purchases per customer
- Conditional aggregates (COUNT with CASE WHEN) and HAVING clauses to calculate seller-level repeat rates while filtering out low-volume sellers

Full SQL Code

All six queries are available in the GitHub repository, with comments explaining the business question each query answers.

Contact

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